



Faculty of Computer Science · Chair of Multimedia Technology · CMS Team Project · Summer Term 2026

AI-Powered Data Storytelling

An Agentic Approach to Moderating the Emotional Tone of Data Narratives

Interim Presentation Report — literature survey, project idea, and work in progress

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Code & data: <https://github.com/memoelsamadony/ai-powered-data-storytelling>

1 Introduction and Motivation

Data storytelling combines data analysis, visualization, and narrative to communicate insights clearly and memorably. LLMs and agentic systems can now generate such stories automatically, but they introduce risks: fluent text is not necessarily faithful, and the way data is framed measurably changes how an audience interprets it. This project surveys AI-driven data storytelling, builds an interactive system, and contributes a capability the literature does not yet address: an **agent that moderates the emotional tone** of a generated data narrative.

The project has five tasks: (a) survey AI/agentic data-story tools; (b) select a dataset with a strong narrative; (c) author a human-written story for it in an interface; (d) generate a story from the same data with a general LLM *plus* an agentic LLM that moderates the emotional content, in the same interface; (e) compare the human and LLM stories via metrics and/or a user study.

2 Literature Survey

A multi-source check confirmed our central observation: **no published system uses an agent to moderate the affective tone of a data narrative** — existing agentic checks target factual correctness only.

Foundations. Segel and Heer [1] define the design space of narrative visualization (genres; the author-driven–reader-driven spectrum). Hullman and Diakopoulos [2] show that editorial framing — including emotional framing — significantly alters interpretation, the **theoretical justification** for an emotional-moderation agent.

LLM and agentic systems. DataNarrative [3] is the closest precedent: one agent generates and a second *verifies at every step*, beating single-LLM baselines. MDSF [4] is an end-to-end multi-agent pipeline with a reusable evaluation kit; InReAcTable [5] builds interactive visual stories from tables via a ReAct loop; and Data Director [6] is an autonomous analyst+designer system that even proposes adding an *evaluation agent* — directly parallel to our idea.

Quality and ethics. Kasner and Dušek [7] report open LLMs are fluent yet historically error-prone at data-to-text — a key trust caveat. DataTales [9] pushes further, into *data narration* (trend,

causal and predictive claims), and shows that even strong models reconstruct a human analyst’s key numbers <30% of the time and reason causally far worse still — evidence that *analytical* claims, not only raw facts, need checking. Lo Duca [8] is the one surveyed framework foregrounding **ethics, misinformation and bias**, reinforcing the need for tone moderation.

3 Surveyed AI Tools and Agentic Systems

System (year, venue)	Type	Role / domain in our context
Voder (2018, IEEE InfoVis)	Template NLG	Interactive data facts from a chart; non-LLM baseline.
Calliope (2021, IEEE TVCG)	Algorithmic (MCTS)	Auto visual data stories from a spreadsheet; non-LLM baseline.
DataNarrative (2024, EMNLP)	LLM, two agents (gen + verify)	Closest precedent; we swap its factual verifier for an emotional moderator.
MDSF (2025, arXiv)	LLM multi-agent pipeline	Pipeline blueprint + ready-made evaluation kit (Task 5).
InReAcTable (2025, UIST)	LLM ReAct agent	Interactive visual stories from tables; UI patterns.
Data Director (2024, IEEE VIS)	LLM multi-agent (analyst+designer)	Autonomous end-to-end animated data video; reusable narration prompts.
DataTales (2025, arXiv)	LLM benchmark (data narration)	Real-world data-narration benchmark; reproduced (partial) — grounds our analytical-correctness metrics (causal ops scored 0%).
Open LLMs: Gemma, Qwen, Zephyr + Opus 4.7 as judge	LLM generators + LLM-as-judge	Our generation backbone and scalable evaluation (see §5).

4 Identified Gap and Our Idea

Across these systems agentic verification always targets *facts*. Yet framing research [2] and the only ethics-focused framework [8] show the *emotional* framing of a data story changes how it is received and can mislead even when every number is correct. Our contribution fills this gap.

Idea. Following the generator–critic pattern of DataNarrative [3] and Data Director [6], we pair a **general LLM that generates** the data story with a second **agentic LLM that moderates its emotional tone** thus detecting manipulative fear, false reassurance or numbing detachment and proposing tone-balanced revisions — while a lightweight factual check (informed by [7]) keeps the content faithful. The same dataset is also given to a human author, and the two stories are compared.

5 Work in Progress

5.1 Reproductions (data-to-text reality check)

We reproduced three prior systems offline to ground our design (open models via Ollama; **Opus 4.7** as the LLM judge).

Kasner and Dušek [7] : faithfulness. Re-running their reference-free error-span method, a modern 12B model (gemma4:12b) had only $\approx 18\%$ of outputs with ≥ 1 semantic error vs the paper’s $>80\%$, while a 4B model regressed to 52% — both size and recency matter. An adversarial second-pass judge confirmed the low rate is genuine, not leniency.

DataTales [9] : data narration (partial). On a 30-report equity slice we reproduced the paper’s **per-operation** complexity gradient and **masked-number** factuality metric: reading/computing operations improve sharply with scale (e.g. rate-of-change 43% $\rightarrow$ 89% from 4B to 12B), but the **causal** operation scored **0% for both models** and masked-number prediction stayed in the paper’s **sub-30%** regime (12.2% / 0.9%) — causal/explanatory claims are a capability wall, not a size problem.

DataNarrative [3] : generate+verify architecture. reproduced its two-agent generate-and-verify pipeline, the structural basis of our system, confirming the second-agent pattern we adapt — swapping

its factual verifier for an emotional-tone moderator.

Takeaway. Modern open LLMs are already fairly faithful at *stating* data (so tone moderation, not another fact-checker, is our novelty), yet they *confabulate causal claims* — which is why a lightweight factual/causal check sits beside the tone agent.

5.2 Datasets and a working pipeline

Because tone failure is **two-sided**, we deliberately use **two datasets**. The **primary** dataset merges **measles cases** with **MCV1 vaccination coverage** (country×year, OWID/WHO/WUENIC; 9,959 rows, 1980–2024) — its natural failure mode is **alarmism**, so the moderator must pull an over-alarmist story *down* without losing real urgency. The **secondary** dataset is the **WHO Global Health Observatory** (child-mortality and life-expectancy trends) — a “hope/progress” story whose failure mode is the **opposite**: over-optimism and false reassurance, so the moderator must *keep the gravity* (the remaining inequality, the COVID-era reversal) rather than flatten it. Together they show the agent calibrates in **both directions** and is **specific** — many edits on the alarmist story, few on the already-measured one.

On the primary data we ran a first pipeline: **qwen3.5:4b generates** a story → **gemma4:12b moderates** its tone → **Opus 4.7 judges**. The small model produced fluent but emotive text and a hallucinated number; the moderator removed the alarmism and, by re-grounding, corrected the number — but did not flag it, confirming the need for a separate factual check beside the tone agent.

5.3 Evaluation metrics

Grounded in the reproductions, our metric set is: **faithfulness**: % of outputs with ≥ 1 semantic error and % of stated numbers correct; **analytical correctness**: per-operation accuracy (lookup, comparison, trend, rate-of-change, causal); **tone calibration (novel)**: an LLM-judge alarmism rating (1–5) measured *before vs after* moderation and the count of emotive spans removed, with faithfulness re-checked after moderation to confirm the edit preserves facts; and a **user study**: on trust, engagement, readability and human-vs-LLM preference.

6 Plan and Next Steps

- Build the interactive web interface hosting both the human and LLM stories.
- Author the human baseline story; add the factual-consistency agent to the pipeline.
- **Scale the models up**: re-run generation and moderation with progressively larger open models (e.g. qwen3.5:4b → 9b, then 12B-class) to measure how model size affects the hallucination rate, the amount of tone moderation required, and the persistent causal-reasoning gap.
- Run the human-vs-LLM comparison on *both* datasets (metrics in §5.3 + a user study).

References

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